OPERATING SYSTEMS

Process Coordination & Deadlocks

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Semaphores

- Semaphore(S) is a counter that is manipulated atomically through two operations : **Signal and Wait.**
- **Wait: decrement if counter is zero** then block until the semaphore is signaled
- **Signal: increment counter**, wakeup one waiter if any
- Operations **`wait()`** and **`signal()`** must be executed atomically
- Semaphore types: Counting Semaphore, Binary Semaphore
- Binary – integer value can be 0 or 1 (known as mutex)
- Counting – integer value can be a range of values

Definition of the **wait()**operation

```
wait(S) {
    while (S <= 0);
     S=s-1; }
```

Definition of the **signal()**operation

```
signal(S) {
    S=s+1;
}
```

Mutex (Mutual exclusion) is a shared variable used by multiple process whose value is only binary as it is a binary semaphore and it is initialized with 1

Hence, the semaphore S in the wait and signal can be replaced by mutex

wait(mutex) { while (mutex <= 0); mutex--; }	**signal(mutex) { mutex++; }**

Semaphores

```
do {
Wait(mutex);
    //critical section
Signal(mutex);
   //remainder section
} while (TRUE);
```

THANK YOU

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